## Question

It is known that the chemical formula of a cluster compound $\mathbf{W}$ containing a p-block non-metallic element can be expressed as $\mathbf{A}_2\mathbf{B}_{14}Cl_{14}$, and its structure contains an electrically neutral fragment of $\mathbf{B}_{10}$ in the middle. The transition metal $\mathbf{A}$ is coordinated by 6 $\mathbf{B}$ atoms.

It is known that 5.346 g of the cluster compound $\mathbf{W}$ can be precisely obtained from 0.768 g of the transition metal element $\mathbf{A}$, 2.688 g of the element $\mathbf{B}$, and a certain amount of chloride $\mathbf{C}$ of $\mathbf{B}$ through a combination reaction at 250 $^{\circ}\mathrm{C}$. If $\mathbf{W}$ is heated to 280 $^{\circ}\mathrm{C}$ in a hydrogen atmosphere, then $\mathbf{W}$ will decompose into the element $\mathbf{B}$, 2.133 g of $\mathbf{D}$, and 0.597 g of $\mathbf{E}$, and 0.4032 L of gas $\mathbf{F}$ (341.1 $^{\circ}\mathrm{C}$, 1 $\mathrm{atm}$). $\mathbf{D}$ is a coordination compound of $\mathbf{A}$, belonging to the $D_{3d}$ point group. After separating the element $\mathbf{B}$ from the above decomposition products, the remaining solid residue is further heated to a higher temperature to obtain 1.792 g of a ternary compound $\mathbf{G}$, and two gases $\mathbf{C}$ and $\mathbf{F}$ are produced in a 1:1 ratio. $\mathbf{C}$ and $\mathbf{F}$ have the same elemental composition, and the volume of the mixed gas at 409 $^{\circ}\mathrm{C}$ and 1 $\mathrm{atm}$ is 0.224 L. The molar mass of $\mathbf{E}$ is 0.28 times that of $\mathbf{D}$. Then the following options are correct.

A. $\mathbf{W}$ decomposition yields chloride $\mathbf{C}$ of $\mathbf{B}$ containing 6 chlorines

B. The molecular weight of $\mathbf{F}$ is higher than that of $\mathbf{C}$

C. $\mathbf{D}$ contains 6 chlorine atoms.

D. $\mathbf{G}$ can be represented as $\mathbf{A}_2\mathbf{B}$

E. 5.346 g $\mathbf{W}$ decomposes to produce 2.04 g of elemental $\mathbf{B}$

F. The ratio of $\mathbf{G}$, $\mathbf{C}$, and $\mathbf{F}$ generated by heating after removing $\mathbf{B}$ is 2:1:1.

G. The ratio of $\mathbf{B}$ and $\mathbf{C}$ generated by the $\mathbf{W}$ decomposition is 2:1

**H.** All of the above options are incorrect.

## Answer

**Correct Answer:** F

## Detailed Explanation

Solution process (Keep the original format, only change the uppercase letters to format):

The reaction of synthesizing $\mathbf{W}$ is a combination reaction. Let the chemical formula of $\mathbf{C}$ be $\mathbf{B}Cl_x$. We can try to write the equation of the reaction:

$$2\mathbf{A} + (14 - 14/x)\mathbf{B} + (14/x)\mathbf{B}Cl_x \rightarrow \mathbf{A_2B_{14}}Cl_{14}$$

Obviously, the mass of $\mathbf{C}$ is $5.346 - 0.768 - 2.688 = 1.890(g)$

So we have:

$$\frac{2.688}{M(\mathbf{B})} \cdot \frac{14}{x} = \frac{1.890}{M(\mathbf{B})+35.45x} \times (14 - 14/x)$$

Simplifying, we get:

$$M(\mathbf{B}) = \frac{35.45x}{0.703(x-1)-1}$$

**CHECKPOINT** 1 PTS

$$M(\mathbf{B}) = \frac{35.45x}{0.703(x-1)-1}$$

Obviously $x > 2$. A simple discussion shows that when $x = 4$, $M(\mathbf{B}) = 127.9$, corresponding to the $Te$ element.

So the molecular formula of $\mathbf{C}$ is $TeCl_4$.

CHECKPOINT 1 PTS

$\mathbf{B}$ is $Te$, $\mathbf{C}$ is $TeCl_4$

Therefore, option $\mathbf{A}$ is incorrect.

At the same time, it is obvious that

$$\frac{0.768}{2M(\mathbf{A})} = \frac{5.346}{2M(\mathbf{A})+35.45\times14+127.9\times4}$$

Solving for $M(\mathbf{A}) = 191.8$, corresponding to the $Ir$ element

So the molecular formula of $\mathbf{W}$ is $Ir_2Te_{14}Cl_{14}$

CHECKPOINT 1 PTS

$\mathbf{A}$ is $Ir$, $\mathbf{W}$ is $Ir_2Te_{14}Cl_{14}$

According to the thermal decomposition experiment of the mixture $\mathbf{D}+\mathbf{E}$, the total mass of $1:1$ $\mathbf{C}$ and $\mathbf{F}$ gases is $m = 2.133 + 0.597 - 1.792 = 0.938(g)$

The average molar mass of the mixed gas of $\mathbf{C}$ and $\mathbf{F}$ is $M = mRT/pV = 234.5\ g\,mol^{-1}$

It is known that $\mathbf{C}$ is $TeCl_4$, $M(\mathbf{C}) = 269.7\ g\,mol^{-1}$. Obviously, $M(\mathbf{F}) = 199.3\ g\,mol^{-1}$, and $\mathbf{F}$ is composed of the same elements as $\mathbf{C}$. It is easy to know that $\mathbf{F}$ is $TeCl_2$

CHECKPOINT 1 PTS

$\mathbf{F}$ is $TeCl_2$

Therefore, option $\mathbf{B}$ is incorrect.

In the decomposition process of $\mathbf{W}$, the mass of $\mathbf{F}$ produced is $m(\mathbf{F}) = pVM/RT = 1.597\ g$

So $m(\mathbf{B}) = 5.346 - 2.133 - 0.597 - 1.597 = 1.019(g)$

CHECKPOINT 0.5 PTS

Mass of the simple substance produced during decomposition $m(\mathbf{B}) = 1.02\,g$

Therefore, option $\mathbf{E}$ is incorrect.

Comparing, we can find that $n(\mathbf{B}) = n(\mathbf{F}) = 4n(\mathbf{W})$

Therefore, option $\mathbf{G}$ is incorrect.

Now try to write the decomposition equation of $\mathbf{W}$. First, assume that $1$ molecule of $\mathbf{D}$ and $\mathbf{E}$ are obtained:

$$Ir_2Te_{14}Cl_{14} \to 4Te + 4TeCl_2 + \mathbf{D} + \mathbf{E}$$

$\mathbf{D}$ and $\mathbf{E}$ add up to $Ir_2Te_6Cl_3$. According to the question, $M(\mathbf{E}) = 0.28M(\mathbf{D})$

We can get $M(\mathbf{E}) = 298.6\ g\,mol^{-1}$, corresponding to $\mathbf{E}$ being $IrCl_3$, so $\mathbf{D}$ is $IrTe_6Cl_3$, which also meets the symmetry of $\mathbf{D}$ in the question, that is, $[IrCl_3]$ is at the center of $[Te_6]$.

**CHECKPOINT** 1 PTS

$\mathbf{E}$ is $IrCl_3$, $\mathbf{D}$ is $IrTe_6Cl_3$

Therefore, option $\mathbf{C}$ is incorrect.

The equation for the thermal decomposition experiment of the $\mathbf{D}+\mathbf{E}$ mixture:

$$IrCl_3 + IrTe_6Cl_3 \rightarrow \mathbf{G} + TeCl_2 + TeCl_4$$

$\mathbf{G}$ is $IrTe_2$, and the reaction equation is:

$$IrCl_3 + IrTe_6Cl_3 = 2IrTe_2 + TeCl_2 + TeCl_4$$

**CHECKPOINT** 1 PTS

The composition of $\mathbf{G}$ is $IrTe_2$

**CHECKPOINT** 1 PTS

$$IrCl_3 + IrTe_6Cl_3 = 2IrTe_2 + TeCl_2 + TeCl_4$$

Therefore, options $D$, $G$, and $H$ are incorrect, and $F$ is correct.